

Toyota Racing Development USA

Summer Internship 2026

Alexander Massoumi

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Introduction



UC Irvine Experience

- Solar Racing (UCI ZotSun Solar Racing)
 - Suspension & Steering Team Lead
- Cold Spray Additive Manufacturing Research

Past Experience

- MD Anderson Cancer Center, Summer 2025
 - Cancer Physics and Engineering Research Intern
 - Computational Model Development

Education

- University of California, Irvine
 - B.S., Mechanical Engineering, Class of 2027
 - B.S., Aerospace Engineering, Class of 2027
 - Minor, Materials Science and Engineering

Hobbies / Fun Facts

- Track Marshal / Volunteering
 - Circuit of the Americas, MotoGP
- Class B License Certified



Material Lab Assistance

- Support daily tasks of the Failure Analysis Group
- Valve Spring Failure Analysis
 - Valve Spring Failure Backup
- Piston Hardness Testing
 - Rockwell (HRE)
- Metallographic Mounting



Valve Spring Failure Analysis

- Macro Photos
 - Macro Camera: **Cannon SX610 HS**
- Optical / Confocal Analysis of Fracture Faces
 - Optical Microscope: **Nikon SMZ800**
 - Confocal Microscope: **Keyence VHX-5000**
- SEM Microscopy Shadowing
 - Lab Specimen Prep
 - Cutting Equipment: **Struers Labotom-3**
 - Demagnetizer: **R. B. Annis Co., Inc.**
 - Ultrasonic Cleaner: **Branson 2510**
- Failure Report Investigation Documentation
 - Basic Information / Surface Initiation Inputs
 - 100+ Open EN Spring Failure Analysis



Input
Mileage at Failure
Max Lift
Cylinder
IN / EX
Outer / Inner
TRD Serial #
Coils from Top
Origin Orientation
Initial Crack Angle
Overall FF Break
Cause
Cause Details
Segments
Upper Fracture Face
Tag Ends Intact?
Both springs broken?
Segment used for findings

Surface Initiation Inputs
Initial Crack Shape
Fatigue Field Plane Angle
SEM examination?
Inclusion Inputs
Fatigue field plane angle
Cavity only?
Inclusion material located?
Inclusion chemistry
Inclusion depth from surface
Inclusion/cavity long dimension
Inclusion/cavity short dimension

Figure 1. Failure Report Builder

Valve Spring Imaging

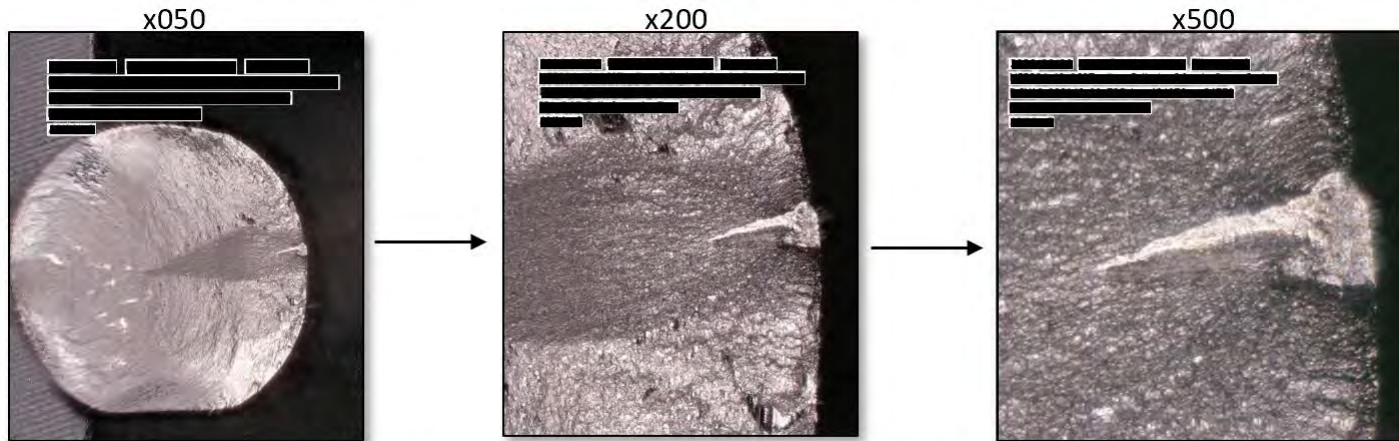


Figure 2. Valve Spring Fracture Face Keyence Imaging

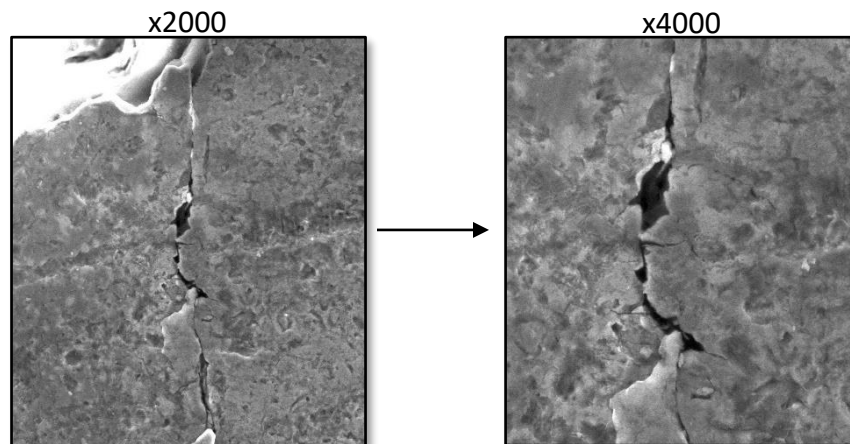
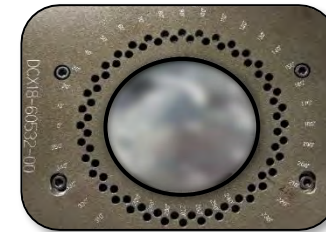


Figure 3. Valve Spring Inner Diameter SEM Imaging

Rockwell Hardness Testing

- Piston Hardness Testing
 - Hardness Tester: United Testing System Tru-Blue II
 - Angle: 0 / 4 / 7 / 8 / 11
 - Orientation: 230 / 250 / 255 / 270 / 290 / 310 / 320
 - Position: Intake / Exhaust / Center

- Hardness Test Documentation
 - Measured hardness reported in excel



Piston #1 (S/N 053)			Piston #3 (S/N 054)			Piston #5 (S/N 055)			Piston #7 (S/N 036)		
Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)	Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)	Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)
	I1				I1				I1		
	I2				I2				I2		
	I3				I3				I3		
	C1				C1				C1		
	C2				C2				C2		
	C3				C3				C3		
	E1				E1				E1		
	E2				E2				E2		
	E3				E3				E3		
Piston #2 (S/N 033)			Piston #4 (S/N 034)			Piston #6 (S/N 055)			Piston #8 (S/N 056)		
Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)	Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)	Notes	Indent	Hardness (HRE)	Hardness (HRB, Converted)
	I1				I1				I1		
	I2				I2				I2		
	I3				I3				I3		
	C1				C1				C1		
	C2				C2				C2		
	C3				C3				C3		
	E1				E1				E1		
	E2				E2				E2		
	E3				E3				E3		

Figure 4. Measured Piston Hardness Excel



Rockwell Hardness Data

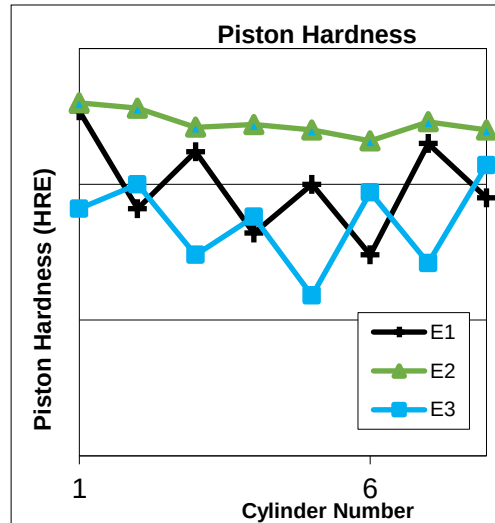
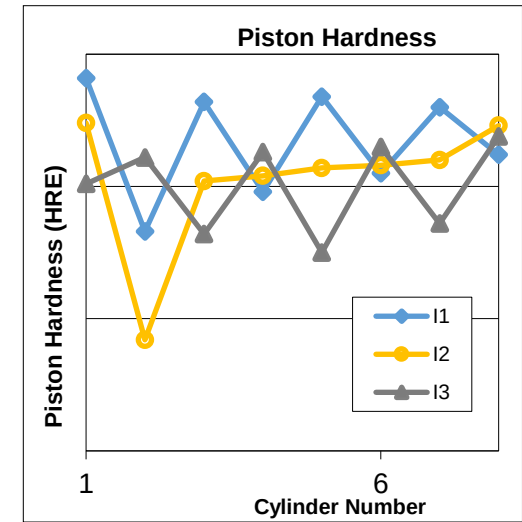
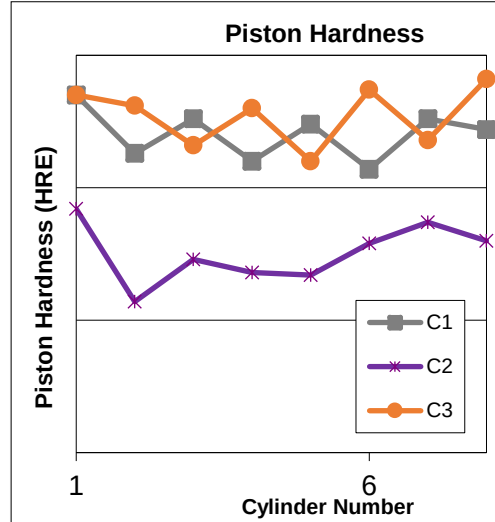
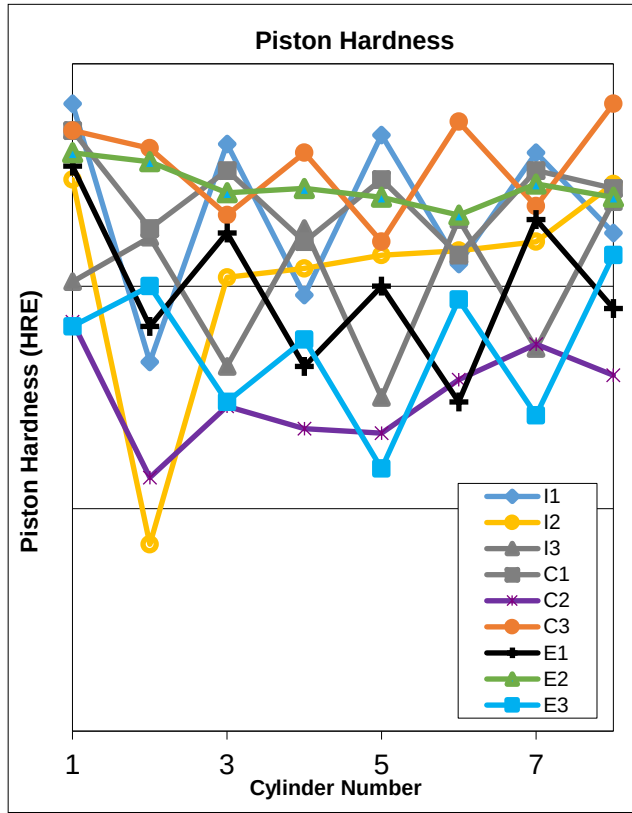


Figure 5. Rockwell Hardness Measurement Plots (HRE)

Rockwell Hardness Data

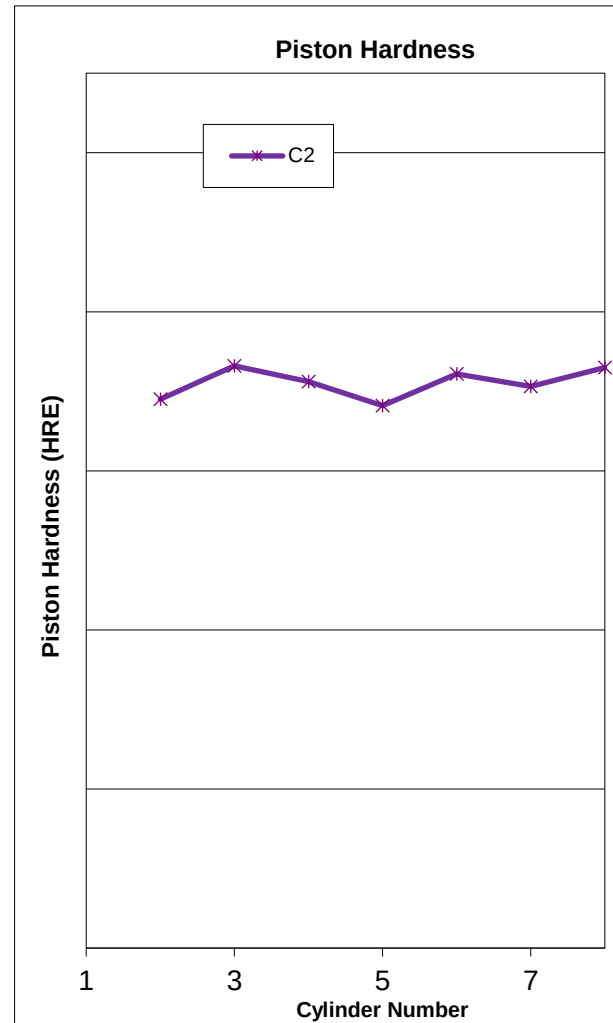


Figure 6. R15 Michigan
Piston Hardness (HRE)

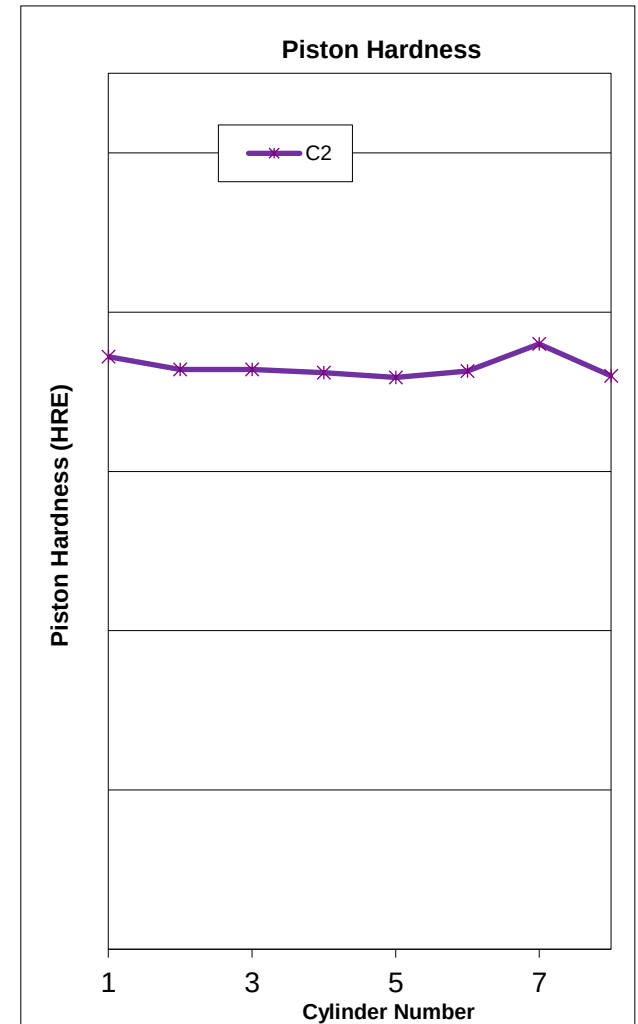


Figure 7. R17 San Diego
Piston Hardness (HRE)

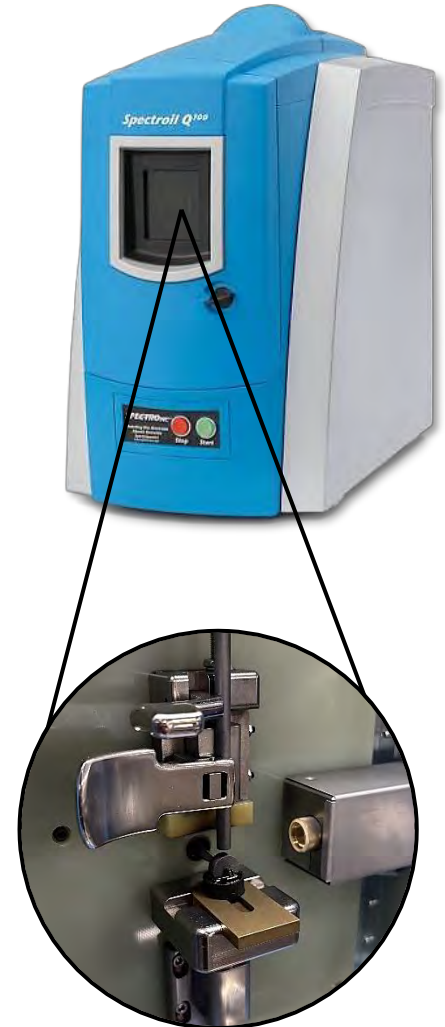
Metallographic Mounting

- Mount Preparation
 - Cutting Equipment: **Struers Labotom-3**
 - Mounting Equipment: **Struers CitoPress-30**
- Material Composition
 - 15 ccm **DuroFast Epoxy Resin**
 - 30 ccm **MultiFast Phenolic Resin**
- Mount Polishing
 - Polishing Machine: **Struers Tegramin-30**



SpectrOil Q100 Elemental Spectrometer Re-Commissioning

- Research, understand, standardize equipment & work process
 - Rotating Disc Electrode Optical Emission Spectroscopy (RDE-OES)
 - Adhere to Spectro Scientific-specific standards & industry standards (ASTM D6595)
 - Accurate, repeatable results
- Create documentation for spectrometer testing
 - Set up necessary equipment
 - Operate initial testing and troubleshooting
 - Standardize work process



Initial Test Runs

- Standardization Oils
 - Base: Standard Oil 75 Base 500mL
 - Lower End: CS-7 100ppm 200G
 - Upper End: SMA-900-200G
- Test Runs
 - 5 Learning Runs to understand the equipment
 - 1 Warm-Up Run to ensure system is warmed up
 - 5 Test Runs for data collection and interpretation
- Side Notes for Skewed Results
 - Graphite Rods and disc electrodes potentially contaminated
 - Standardization Oils – additive separation



Standardization Test Runs

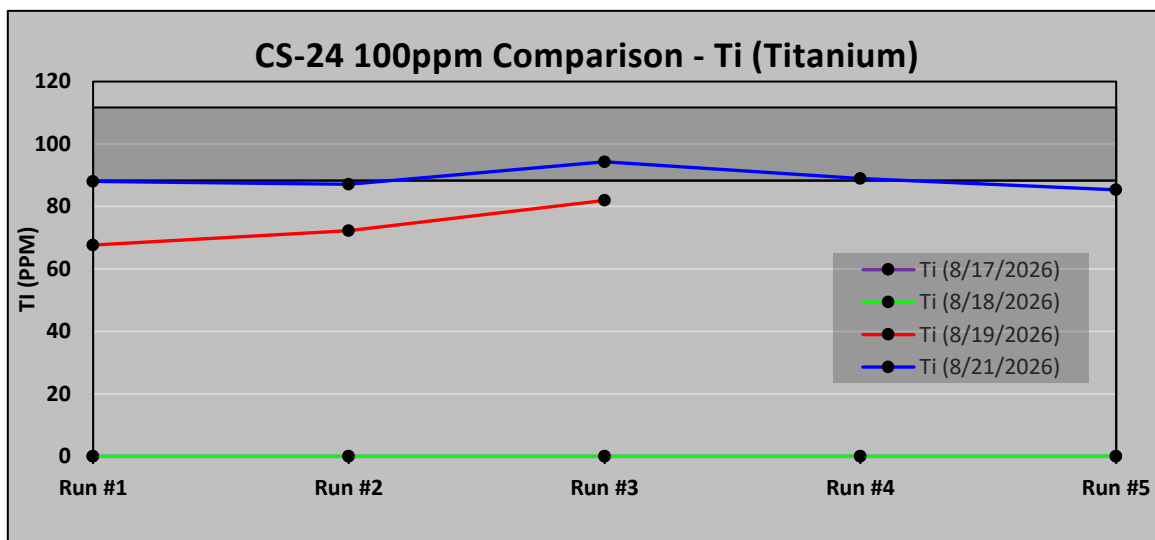
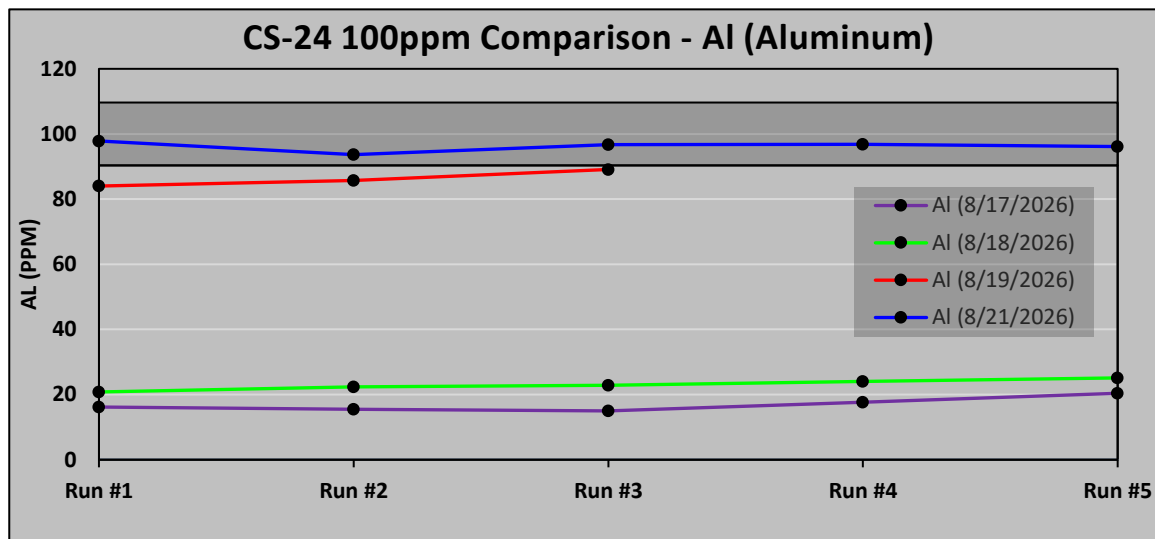
- Standardization Oils
 - Base: Standard Oil 75 Base 500mL
 - Lower End: CS24 100ppm 200G
 - Upper End: SMA-900-200G
- Calibration Process
 - Disc Electrode Offset Procedure – **5 Runs** of 0ppm standardization oil
 - Optical Profile – **1 Run** of CS24 100ppm standardization oil
 - Re-Standardization – **5 Runs** of each standardization oil
- Test Runs
 - **3-5 Warm-Up Runs** to ensure system is warmed up
 - **3 Test Runs** for data collection and interpretation
 - **1 Run** of base oil between each set of test runs to avoid cross-contamination/carryover



Results – sn0000544

Conc, 100ppm	Aluminum (Al)					Mean
Pre	16.21	15.53	14.99	17.64	20.40	16.95
Pre	20.80	22.37	22.85	24.05	25.11	23.04
During	84.01	85.70	89.10			86.27
During	97.84	93.65	96.76	96.79	96.17	96.24

Table 1. Experimental Data for Aluminum (Al)
CS-24-100-200G – Mean



Conc, 100ppm	Titanium (Ti)					Mean
Pre	0.00	0.00	0.00	0.00	0.00	0.00
Pre	0.00	0.00	0.00	0.00	0.00	0.00
During	67.7	72.3	82.0			74.0
Post	88.0	87.1	94.3	89.0	85.3	88.7

Table 2. Experimental Data for Titanium (Ti)
CS-24-100-200G – Mean

Results – sn0000544 cont.

Conc, 100ppm	Aluminum (Al)					Mean	Std. Dev.
Pre	16.21	15.53	14.99	17.64	20.40	16.95	2.17
Pre	20.80	22.37	22.85	24.05	25.11	23.04	1.64
During	84.01	85.70	89.10			86.27	2.59
Post	97.84	93.65	96.76	96.79	96.17	96.24	1.57

Table 3. Experimental Data for Aluminum (Al)
CS-24-100-200G – Mean + Std. Dev.

Conc, 100ppm	Titanium (Ti)					Mean	Std. Dev.
Pre	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pre	0.00	0.00	0.00	0.00	0.00	0.00	0.00
During	67.7	72.3	82.0			74.0	7.30
Post	88.0	87.1	94.3	89.0	85.3	88.7	3.39

Table 4. Experimental Data for Titanium (Ti)
CS-24-100-200G – Mean + Std. Dev.



Aluminum (Al)

- Spectro Scientific Acceptable Mean – 100 ± 9.65
- Spectro Scientific Acceptable Std. Dev. – 5.03
- ASTM D6595 Acceptable Std. Dev. – 4.3



Titanium (Ti)

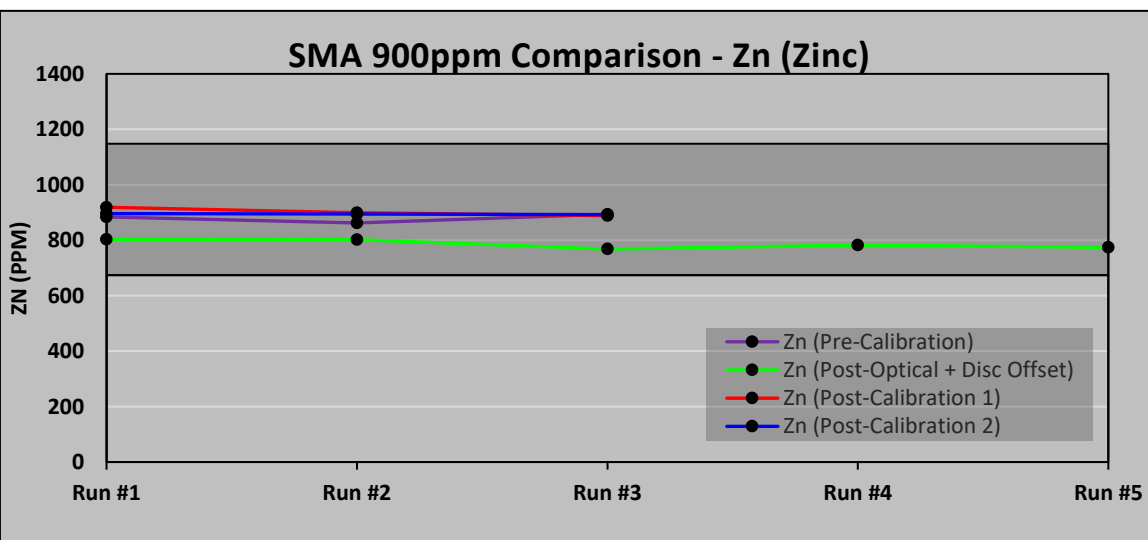
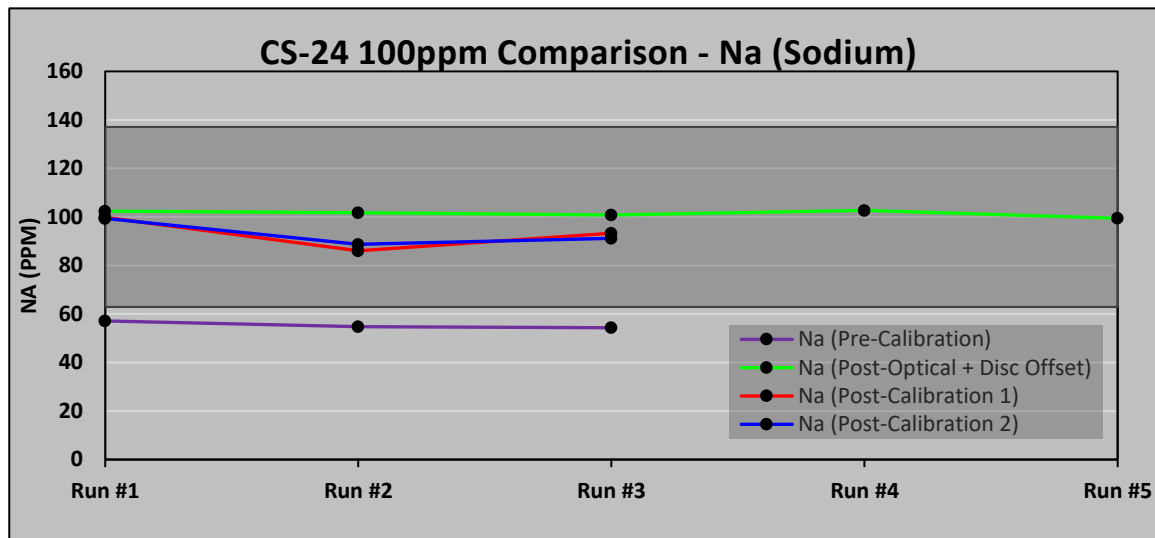
- Spectro Scientific Acceptable Mean – 100 ± 11.7
- Spectro Scientific Acceptable Std. Dev. – 6.02
- ASTM D6595 Acceptable Std. Dev. – 4.2



Results – sn0000543

Conc, 100ppm	Sodium (Na)					Mean
Pre	57.1	54.8	54.3			55.4
During	102.4	101.8	100.8	102.7	99.4	101.4
Post 1	99.8	86.1	93.3			93.1
Post 2	99.4	88.7	91.1			93.1

Table 5. Experimental Data for Sodium (Na)
CS-24-100-200G – Mean



Conc, 100ppm	Zinc (Zn)					Mean
Pre	884.1	862.2	893.7			880.1
During	802.8	802.3	768.2	783.0	774.3	786.1
Post 1	918.9	898.7	888.5			902.0
Post 2	896.1	894.0	892.4			894.2

Table 6. Experimental Data for Zinc (Zn)
SMA-900-200G – Mean

Results – sn0000543 cont.

Conc, 100ppm	Sodium (Na)					Mean	Std. Dev.
Pre	57.1	54.8	54.3			55.4	1.51
During	102.4	101.8	100.8	102.7	99.4	101.4	1.34
Post 1	99.8	86.1	93.3			93.1	6.89
Post 2	99.4	88.7	91.1			93.1	5.59

Table 7. Experimental Data for Sodium (Na)
CS-24-100-200G – Mean + Std. Dev.

Conc, 100ppm	Zinc (Zn)					Mean	Std. Dev.
Pre	884.1	862.2	893.7			880.1	16.1
During	802.8	802.3	768.2	783.0	774.3	786.1	15.9
Post 1	918.9	898.7	888.5			902.0	15.5
Post 2	896.1	894.0	892.4			894.2	1.85

Table 8. Experimental Data for Zinc (Zn)
SMA-900-200G – Mean + Std. Dev.



Sodium (Na)

- Spectro Scientific Acceptable Mean – 100 ± 37.1
- Spectro Scientific Acceptable Std. Dev. – 8.02
- ASTM D6595 Acceptable Std. Dev. – 12.0



Zinc (Zn)

- Spectro Scientific Acceptable Mean – 900 ± 226
- Spectro Scientific Acceptable Std. Dev. – 108
- ASTM D6595 Acceptable Std. Dev. – 82.9



Documentation

- Standardized Work Instructions
 - Developed spectrometer SOP

Project (1)

- Failure Analysis - Fatigue
- Visual / Testing Equipment
 - Keyence, United Hardness Tester
 - Metallographic Mount Equipment

Project (2)

- Planning / Testing
- RDE-OES Testing
 - Engine Oil

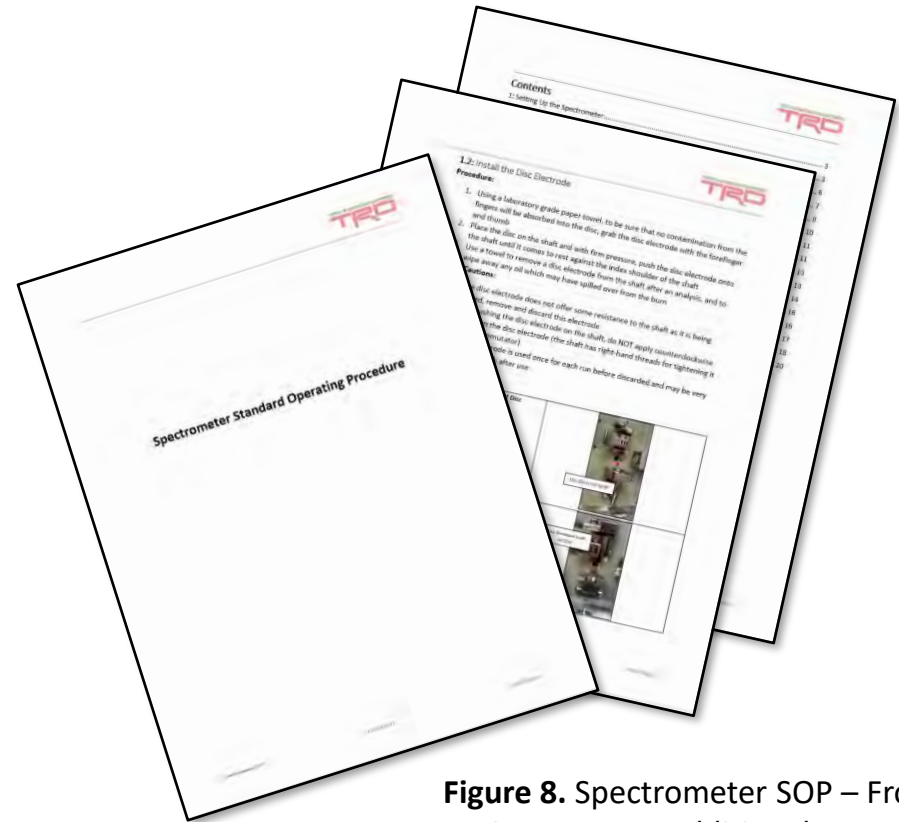


Figure 8. Spectrometer SOP – Front Cover + Two Additional Pages

ASTM D6595 Repeatability – Std. Dev. Tables

Element	Range, mg/kg	Repeatability, mg/kg
Aluminum	0.23–101	$0.5419(X+0.57)^{0.45}$
Barium	28–115	$0.0694X^{1.18}$
Boron	0.14–120	$0.4280(X+0.1028)^{0.56}$
Calcium	3.7–11460	$0.1106(X+2.184)$
Chromium	0.18–152	$0.7285(X+0.0557)^{0.41}$
Copper	0.47–100	$0.1631(X+0.3459)^{0.85}$
Iron	4.8–210	$0.3159(X+0.0141)^{0.73}$
Lead	0.43–101	$1.062(X+0.6015)^{0.34}$
Magnesium	4.9–1360	$0.1049X$
Manganese	0.3–117	$0.7017(X+0.3534)^{0.34}$
Molybdenum	0.21–100	$0.9978(X+0.4795)^{0.34}$
Nickel	0.35–100	$0.7142(X+0.3238)^{0.40}$
Phosphorus	52–2572	$0.0761(X+14.76)$
Potassium	0.35–247	$0.4075(X+0.1154)^{0.63}$
Silicon	3.2–142	$0.4015(X+0.1692)^{0.63}$
Silver	31–102	$0.1523(X+1.2)^{0.88}$
Sodium	3.6–99.6	$0.1231(X-2.674)$
Tin	30–139	$0.6777(X+0.6578)^{0.45}$
Titanium	6.8–103	$0.5831(X+0.9304)^{0.5}$
Vanadium	2.1–101	$0.6389(X+0.8418)^{0.41}$
Zinc	5.3–1345	$0.2031(X+1.553)^{0.87}$

X = mean concentration in mg/kg

Table A.1. Repeatability

Element	mg/kg			
	1	10	100	1000
Aluminum	0.7	1.6	4.3	
Barium			16.2	
Boron	0.5	1.6	5.6	
Calcium		1.3	11.3	110.8
Chromium	0.7	1.9	4.8	
Copper	0.2	1.2	8.2	
Iron		1.7	9.3	
Lead	1.2	2.4	5.1	
Magnesium		1.0	10.5	104.9
Manganese	0.8	1.6	3.4	
Molybdenum	1.1	2.2	4.8	
Nickel	0.8	1.8	4.5	
Phosphorus			1.9	8.7
Potassium	0.4	1.8	7.4	
Silicon		1.7	7.3	
Silver			8.9	
Sodium		0.9	12.0	
Tin			5.4	
Titanium		1.7	4.2	
Vanadium		1.7	4.2	
Zinc		1.7	11.3	82.9

Table A.2. Calculated Repeatability at Selected Concentrations in mg/kg

SpectrOil Q100 Oil Analysis Spectrometer Operator and User Maintenance Guide Tables

Conc, PPM	Al Cr Ni Si	Ti B	Fe Ag Mo	Cu Mg	Pb Sn	Zn	Na	V Mn Cd Ba	Ca	P
0	0.88	0.89	0.91	0.92	1.60	0.96	1.01	0.5	0.5	N/A
5	1.20	1.30	1.50	1.61	1.98	1.99	2.59	N/A	N/A	N/A
10	1.59	1.78	2.21	2.44	2.43	3.19	4.36	1.5	1.5	N/A
30	3.33	3.93	5.23	5.91	4.47	8.15	11.6	3.25	3.25	15.5
50	5.12	6.14	8.29	9.43	6.64	13.1	18.9	5.5	5.5	18.2
100	9.65	11.7	16.0	18.2	12.2	25.6	37.1	10	10	20
300	27.8	33.9	46.7	53.5	34.3	75.6	110	32	32	35
500	46.0	56.1	77.5	88.8	56.6	126	183	53	53	60
700	64.2	78.3	108	124	78.8	176	255	N/A	N/A	N/A
900	82.4	101	139	159	101	226	328	95	95	105
5000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	500	500
10000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1000	1000

Table B.2. Acceptable Accuracy Means for Wear Materials - Mean

Conc. PPM	Fe, Al, Cr, Cu, Mg, Ni, Si	Ti, B	Ag, Na, Mo	Pb	Sn	Zn	V, Mn, Cd, Ca, Ba	P
0	0.50	0.50	0.50	0.90	1.00	0.50	0.5	N/A
5	0.56	0.58	0.64	0.95	1.04	0.78	N/A	N/A
10	0.71	0.78	0.94	1.08	1.17	1.30	0.75	N/A
30	1.58	1.87	2.45	2.01	2.06	3.63	1.75	5.5
50	2.55	3.04	4.03	3.13	3.16	6.02	2.75	7.5
100	5.03	6.02	8.02	6.07	6.08	12.0	3.75	10

Conc. PPM	Fe Ag Mo	Al Cr Ni Si	Cu Mg	Na	Pb Sn Ti B	Zn	V	Mn	Cd Ba	Ca	P
300	24.0	15.0	27.0	48.0	18.0	36.0	10	12	10	10	15
500	40.0	25.0	45.0	80.0	30.0	60.0	20	25	20	20	25
700	56.0	35.0	63.0	112	42.0	84.0	N/A	N/A	N/A	N/A	N/A
900	72.0	45.0	81.0	144	54.0	108	45	40	40	40	40
5000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	150	200
10000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	300	300

Table B.2. Repeatability Standard Deviations for Wear Materials

- Materials Lab: Geoff, Jason, Guillermo
- HR/Recruiting: Vasti, Laiza, Maegan
- Personnel involved in SoCal Toyota site tours
 - TLS, TMNA R&D Gardena, TABC, LAPDC
- Everyone at TRD

Thank you for an amazing experience and summer!

Questions?